# Rezaul Islam

Dhaka-1000, Bangladesh | +880 1521 237290 | Email: rezaul90@iwfm.buet.ac.bd

## Academic Credentials

**M.Sc. in Civil & Structural Engineering** (Ongoing)

Bangladesh University of Engineering and Technology (BUET), Dhaka, Bangladesh

**B.Sc. in Civil Engineering** (2023)

Bangladesh University of Engineering and Technology (BUET), Dhaka, Bangladesh

- Major: Structural Engineering
- CGPA: 3.89 / 4.00; class rank 9$^{th}$ of 206 (top 4.4%)

## Research Interests

- Coastal engineering, wave mechanics, and wave-structure interaction
- Computational fluid dynamics (CFD) and numerical modeling
- Coastal hydraulics, hydraulic structures, and coastal protection infrastructure
- Hydrological modeling
- Remote sensing and GIS applications
- Structural mechanics and finite element analysis

## Professional Experience

**Lecturer** (December 2023 - Present) | Institute of Water and Flood Management (IWFM), Bangladesh University of Engineering and Technology (BUET)

## Research Experience

**M.Sc. Thesis**

Supervisor: Dr. Raquib Ahsan, Professor, Department of Civil Engineering, BUET

Co-Supervior: Dr. Mohammad Asad Hussain, Institute of Water and Flood Management (IWFM), BUET

Topic: Numerical Study of Seawall Geometries for Enhanced Structural Performance under Hydrodynamic Wave Action

**Undergraduate Thesis**

Supervisor: Dr. Tahsin Reza Hossain, Department of Civil Engineering, BUET

- Topic: Performance Evaluation of RC Moment Frames Using Nonlinear Pushover Analysis

## Research Projects

**Co-Investigator**, Implementation of Geospatial Technique in Flood and Drought Monitoring for the Safety of Nuclear Installations (November 2023 - June 2027)

- Funded by the International Atomic Energy Agency (IAEA); lead: Bangladesh Atomic Energy Commission (BAEC); affiliate: IWFM, BUET

**Co-Investigator**, Anticipatory Action for Health in a Changing Climate (November 2024 - April 2026)

- Funded by Tufts University, USA and The Rockefeller Foundation, USA

**Co-Investigator**, Application and Updating of the CCM for Cyclone Impact Prediction and the DFRM for the Risk-based Riverine Flood Warning in Bangladesh (April 2025 - December 2025)

- Funded by Eco-Social Development Organization (ESDO)

**Co-Investigator**, Enhancing Climate Resilience of Coastal Areas of Bangladesh with Nature Based Solutions (October 2022 - June 2025)

- Funded by the Global Center on Adaptation (GCA); lead: IWFM, BUET

## Advisory Services

**Team Member**, Hydro-Morphology Study of Bridges under Dhaka and Mymensingh Divisions (October 2023 - September 2025)

- Client: Local Government Engineering Department (LGED)

## Publications

Conference

Tushar, M. N., **Islam, R.,** Muktadir, H. M., Shampa, M., Rahman, M. M., Islam, A. K. M. S., & Hussain, M. A. (2025). Assessing the replicability of bandal-like structures for sediment management in the Kuakata region of the coastal zone of Bangladesh. Proceedings of the 10th International Conference on Water and Flood Management (ICWFM 2025), Dhaka, Bangladesh.

**Islam, R.**, Tushar, M. N., Dey, S., Uddin, S. I., Shampa, M., Chowdhury, A. I. A., Murshed, S. B., Haque, S., & Salehin, M. (2025). Predicting Ambient Air Temperature of Bangladesh Using Machine Learning and Remote Sensing. AGU Fall Meeting Abstracts, 2025, A31G-VR8865. Poster link.

Billah, M. M., Sakib, M. R., & **Islam, R.** (2025). Multi-layer perceptron Markov chain-based artificial neural network for modeling future land-specific carbon emission patterns and its effects on land surface temperature in Rajshahi City. AGU Fall Meeting Abstracts, 2025, GC41N-VR8883.

## Technical Skills

- Structural Analysis and Design: ANSYS Mechanical (Static Structural, Transient Structural), ETABS, SAP2000
- Computational Fluid Dynamics (CFD): ANSYS Fluent, CFD-Post
- GIS and Remote Sensing: ArcGIS, raster analysis, GIS-based spatial analysis
- Hydrological Modeling: HEC-HMS
- Programming and Scientific Computing: Python, C++, MATLAB, Mathematica
- Machine Learning and Data Analysis: Machine learning, Python-based data analysis, Microsoft Excel
- Climate and Geospatial Data Processing: NetCDF processing, climate data analysis, geospatial data processing

## Involvements

Co-Instructor: Post graduate seminar course for APRIL-2025 Semester, IWFM, BUET

Assistant Organizing Secretary: International Conference on Water and Flood Management-2025, IWFM, BUET

## Honors and Awards

- University Merit Scholarship, BUET, for academic excellence in undergraduate studies
- Dean's List Award, Department of Civil Engineering, BUET, for academic excellence across all four undergraduate levels (I-IV)